Final Project: The Human Costs of the Covid Lockdown

This project replaces the final exam and counts for **30 points**.

You may discuss ideas with classmates, but **all submitted work must be written independently**. Your explanations and code must be your own.

Submit a Jupyter notebook containing your code, figures, and written explanations.

This project integrates several ideas from the course. You will use macroeconomic data to construct a Markov model of economic states, derive asset prices using a stochastic discount factor, simulate the dynamics of the economy under alternative policies, and evaluate those policies using both economic welfare criteria and ethical reasoning. The final section asks you to combine economic analysis with evidence from epidemiology and moral philosophy to evaluate pandemic policies.

0. Import the Data

Download the **Weekly Economic Index (WEI)** from the FRED database.

Download the series using the **maximum date range available**. Do not truncate the sample; use the entire dataset provided by FRED.

Important: The WEI is reported in percent. Convert the data to decimal form before using it in calculations. For example,

$$1.00\% = 0.01.$$

Be careful to divide the data by 100 before using it in any calculations.

1. Discretizing the State Space (2 points)

Plot the time series of the Weekly Economic Index.

Then construct a histogram with **20 bins**. Interpret each bin as a **state of the economy**.

Label the states

$$s = 1, 2, \ldots, 20.$$

State 1 corresponds to the **lowest bin**, which represents the Covid collapse state.

This bin contains only **six observations**.

We use 20 bins to preserve a distinct extreme lower-tail Covid state, even though this produces relatively few observations in that state. This illustrates the tradeoff between economic resolution and statistical precision when discretizing a continuous variable.

2. Classifying the States (2 points)

Using the bin boundaries from your histogram, assign each weekly observation of the WEI to one of the 20 states.

Let

$$s_t \in \{1, 2, \ldots, 20\}$$

denote the state of the economy in week t.

Your classification must satisfy the following conditions:

- Each observation must belong to **exactly one state**.
- The states must correspond exactly to the **bin boundaries used in your histogram**.
- State 1 is the lowest WEI bin, the Covid collapse state.

Construct and report the time series

$$(s_1, s_2, \ldots, s_T).$$

Finally, report the **number of observations in each state**. Your counts should match the frequencies shown in your histogram.

3. Estimating the Markov Matrix (2 points)

Using the state sequence $(s_1, s_2, \ldots, s_T)$ constructed in the previous problem, estimate the 20×20 Markov transition matrix

$$P_{ij} = \Pr(s_{t+1} = j \mid s_t = i).$$

Compute the transition counts

$$N_{ij} = \#\{t : s_t = i \text{ and } s_{t+1} = j\}.$$

Let

$$N_i = \sum_{j=1}^{20} N_{ij}$$

denote the total number of times the economy is observed in state i and then transitions to the next period.

Estimate the Markov matrix as

$$P_{ij} = \frac{N_{ij}}{N_i}.$$

Your estimated matrix should satisfy

$$\sum_{j=1}^{20} P_{ij} = 1$$

for every state i.

Next compute the **ergodic distribution** π, defined by

$$\pi P = \pi, \qquad \sum_{i=1}^{20} \pi_i = 1.$$

You may compute the ergodic distribution either by

- finding the **left eigenvector** of P associated with eigenvalue 1 and normalizing it so the elements sum to one, or
- iterating an arbitrary probability vector π_0 according to $\pi_{t+1} = \pi_t P$ until it converges.

Report the **long-run probability of being in the Covid state**, which is π_1.

4. Constructing Consumption (2 points)

Construct a consumption level for each state as follows.

For each state i, compute the **average value of the WEI for all observations that fall into that bin**. Let this average be

$$x_i.$$

Do **not** use the midpoint of the histogram bin; use the sample average of the observations classified in that state.

Define the consumption level in state i as

$$c_i = e^{x_i}.$$

Exponentiating the WEI converts the index into a level variable that we interpret as consumption.

The resulting vector

$$c = (c_1, c_2, \ldots, c_{20})$$

is the consumption vector of the economy.

5. Asset Pricing Kernel (4 points)

The representative agent has the constant relative risk aversion (CRRA) felicity function

$$u(c) = \frac{c^{1-\gamma} - 1}{1 - \gamma}.$$

Assume the coefficient of relative risk aversion is

$$\gamma = 2$$

and the discount factor is

$$\beta = 0.97^{1/52}.$$

The marginal utility of consumption is therefore

$$u'(c) = c^{-\gamma}.$$

Define the **asset pricing kernel** (or stochastic discount factor)

$$M_{ij} = \beta \frac{u'(c_j)}{u'(c_i)},$$

which represents the marginal rate of substitution between consumption in state i today and consumption in state j next period.

Using your consumption vector from the previous problem, construct the 20×20 matrix

$$M = (M_{ij}).$$

Explain the following:

- Why are the diagonal elements M_{ii} all equal?
- Why is the matrix M generally **not symmetric**?
- When pricing assets in this economy, why must one use both the **pricing kernel** M and the **Markov transition matrix** P?

6. Yield Curves (2 points)

Consider zero-coupon bonds with maturities

$$[1, 2, 4, 8, 20, 40, 80, 120]$$

quarters.

Because the data are weekly, one quarter corresponds to

$$13 \text{ weeks.}$$

Thus a maturity of k quarters corresponds to $13k$ weeks.

Let $q_i^{(n)}$ denote the price in state i of a zero-coupon bond that pays one unit of consumption in n weeks.

Bond prices satisfy the recursion

$$q_i^{(n+1)} = \sum_{j=1}^{20} P_{ij} M_{ij} q_j^{(n)},$$

with

$$q_i^{(0)} = 1.$$

For each maturity n, convert the bond price into a yield

$$y_i(n) = -\frac{1}{n} \log q_i^{(n)}.$$

Construct the yield curve in each state.

Your figure should contain **20 yield curves**, one for each state of the economy. Each curve should plot the yield $y_i(n)$ against maturity.

7. Yield Curve Inversion (2 points)

An **inverted yield curve** occurs when short-term interest rates exceed long-term interest rates.

Using your model, discuss whether an inverted yield curve conveys information about the **next period's state of the economy**.

In your discussion, refer to:

- the transition probabilities in the Markov matrix P,

- the role of marginal utility in the asset pricing kernel M,
- and the relationship between bond prices and expectations of future economic states.

Explain why the yield curve may become inverted when the economy is expected to transition toward states with high marginal utility of consumption.

8. Pricing the Stock Market (2 points)

Suppose the stock market is a **Lucas tree** that pays a dividend equal to aggregate consumption each quarter.

Thus, if the economy is in state i, the dividend paid during the quarter is

$$d_i = c_i.$$

Let p_i denote the price of the stock market when the economy is in state i.

Asset prices satisfy the recursion

$$p_i = \sum_{j=1}^{20} P_{ij} M_{ij} \left(d_j + p_j \right).$$

Compute the stock market price in each state.

Report the **price-dividend ratio**

$$\frac{p_i}{d_i}$$

for each state of the economy.

Hint: Let A denote the matrix with elements

$$A_{ij} = P_{ij} M_{ij}.$$

Then the pricing recursion can be written in vector form as

$$p = A(d + p).$$

Rearrange this expression to solve the linear system

$$(I - A)p = Ad.$$

9. Duration of a Lockdown (2 points)

Suppose the first row of the Markov matrix is

$$\left(\frac{5}{6}, \frac{1}{6}, 0, \ldots, 0\right).$$

Interpret state 1 as the **Covid lockdown state**. Thus, when the economy is in state 1, it remains in that state next period with probability 5/6.

- What is the **expected duration** of a lockdown implied by this transition probability?

Now modify the first row of the transition matrix so that it becomes

$$(p, 1-p, 0, \ldots, 0), \qquad 0 < p < 1.$$

- Choose p so that the expected duration of the lockdown is **52 weeks**.
- Using this modified transition matrix, recompute the **ergodic distribution**.
- Report the new **long-run probability of being in the Covid state**.

10. Simulation (2 points)

Simulate the economy for **1300 weeks** (approximately 25 years) under two scenarios:

1. the **status quo transition matrix** estimated earlier,
2. the **modified transition matrix** with a 52-week lockdown constructed in the previous problem.

For each simulation:

- Draw the initial state s_0 from the **ergodic distribution**.
- Simulate the sequence of states

  $$s_0, s_1, \ldots, s_{1300}$$

  using the appropriate Markov transition matrix.

- Construct the corresponding time series of **consumption**

$$c_{s_t}$$

and **stock market prices**

$$p_{s_t}.$$

Plot the simulated time series of consumption and the stock market.

To display the two series on a comparable scale, multiply consumption by **1500** before plotting.

Your figure should show the dynamics of the economy under both scenarios.

11. Welfare and Policy Evaluation (8 points)

In this problem you will evaluate the economic and social consequences of a one-year lockdown.

Economic Welfare

Compute the welfare cost of an immediate 52-week lockdown.

Express the welfare loss as the **percentage decrease in long-run average consumption** when comparing the lockdown economy with the status quo economy.

You may also consider a **Rawlsian welfare criterion**, which places particular emphasis on the probability of being in the worst state of the economy.

Epidemiological Outcomes

Use Thomas Sargent's SEIR simulation code to evaluate whether a one-year lockdown reduces deaths from Covid in the long run.

Calibrate the model so that total deaths match the approximate U.S. total of

1.1 million deaths.

Compare the long-run outcomes of the epidemic under

- the status quo policy and
- an immediate one-year lockdown.

Interpreting the Value of Lives Saved

When evaluating policies that extend life for elderly individuals, economists often use the value of a **statistical life-year** (VSLY) or a **quality-adjusted life year** (QALY) rather than the value of an entire statistical life.

Typical estimates used in policy analysis range from roughly $100,000 to $300,000 per life-year.

Two empirical facts about Covid mortality are relevant when interpreting the benefits of lockdown policies:

- Covid mortality was **extremely concentrated among the elderly**. In the United States the median age of death was roughly 78–80 years.
- A large fraction of Covid deaths occurred among residents of **long-term care facilities**, whose remaining life expectancy is often relatively short.

Because of these facts, some economists evaluate the benefits of Covid policies in terms of **life-years saved** rather than the number of deaths avoided.

Ethical Considerations

Discuss whether a year-long lockdown would be socially desirable when both economic welfare and mortality outcomes are considered.

In your discussion, refer to the theory of justice developed by **John Rawls**. In particular, consider the **veil of ignorance**: imagine that individuals must choose social policies without knowing their age, health status, or economic position in society.

Would such individuals prefer the lockdown policy or the status quo?

You may find the following sources helpful.

References

- Briggs, Andrew H., Jason R. Goldstein, Kirsi M. L. A. Kirwin, and others (2021). "Estimating the Quality-Adjusted Life-Year Losses Associated with COVID-19 Deaths." *Health Economics*, 30(4), 699–706.
- Flaxman, Seth, Swapnil Mishra, Axel Gandy, H. Juliette T. Unwin, Thomas A. Mellan, Helen Coupland, and others (2020). "Estimating the Effects of Non-Pharmaceutical Interventions on COVID-19 in Europe." *Proceedings of the National Academy of Sciences*, 117(36), 22035–22044.

- Ioannidis, John P. A. (2021). "Reconciling Estimates of Global Spread and Infection Fatality Rates of COVID-19." *European Journal of Clinical Investigation*, 51(5).
- Rawls, John (1971). *A Theory of Justice*. Cambridge, MA: Harvard University Press.
- Viscusi, W. Kip (2018). *Pricing Lives: Guideposts for a Safer Society*. Princeton, NJ: Princeton University Press.
- Viscusi, W. Kip and Joseph E. Aldy (2003). "The Value of a Statistical Life: A Critical Review of Market Estimates Throughout the World." *Journal of Risk and Uncertainty*, 27(1), 5–76.

Your answer should integrate quantitative evidence from your simulations with a clear economic and ethical argument.